

Data Analysis Workflow

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Workflow, Tools and Skills in Statistical Modelling



Proposed workflow

1. Read
2. Explore
3. Assume
4. Analyse
5. Assess
6. Present
7. Publish

Tools

- Workflow gets more complicated
- Identify challenges at each step of the workflow
  - Data gets bigger
  - Data comes in various formats
  - Data from more than one centres
  - Analysis is becoming more complex
- Not using tools will harm you



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Tools: Identify available tools to maximize efficiency at each workflow

- Look at workflow
- For getting data
- For sharing analysis
- For sharing data
- For reproducibility



Skills

- Data modellers need both knowledge and skills for each step of the workflow
- Skills to read clients or customers
- Skills to simplify
- Skills to convince (communicate findings) audience



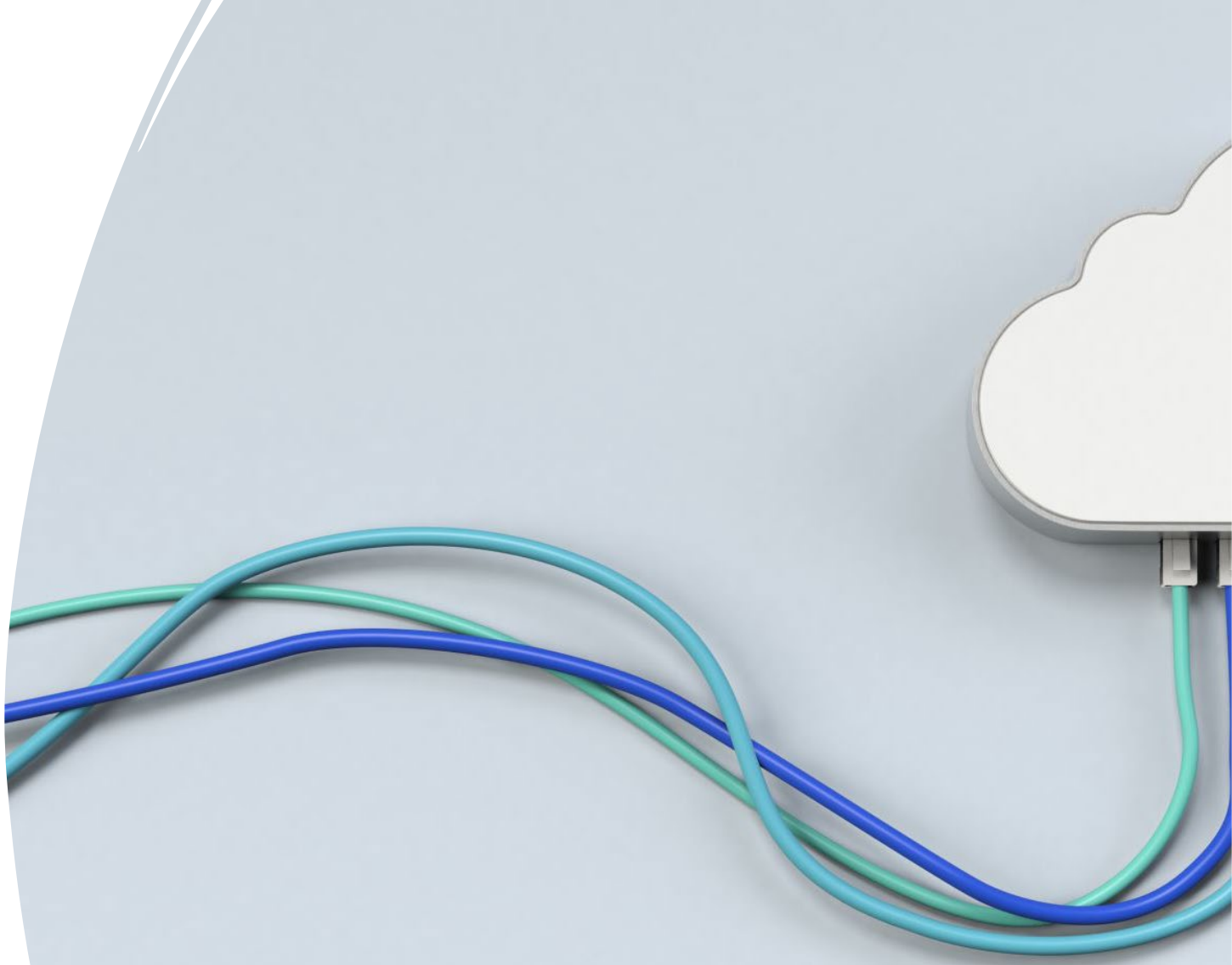
ASK YOURSELF for each of the coming slide….

- Am I using the right tools for my data analysis and modelling?
  - YES
  - NO
  - If NO, what should I do?
- How good am I with the tools?
  - Novice
  - Intermediate
  - Advance



Read data

- Sources of data
  - Offline
  - On the cloud
  - Hybrid
- Structured vs unstructured data
- Common structured data
  - MS Excel
  - Comma separated values
  - Statistical software data
    - SPSS
    - SAS
    - Stata



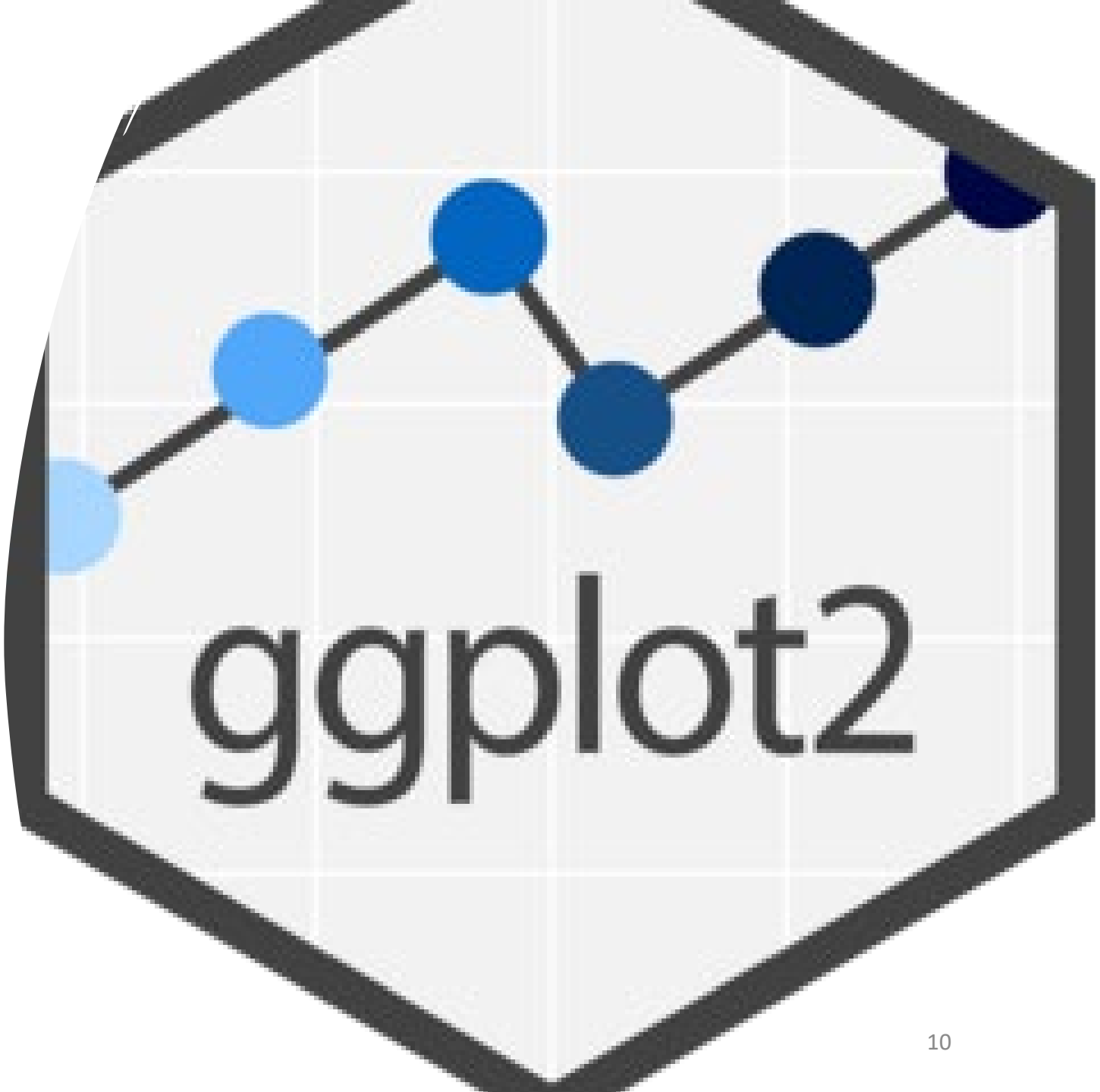
Explore data

- Types
  - Visualization
  - Tabulation
- Do not miss this stage
- Objectives of data exploration
  - In statistics, we use exploratory data analysis (EDA) to summarize main characteristics
  - We use statistical graphics and other data visualization methods or tables



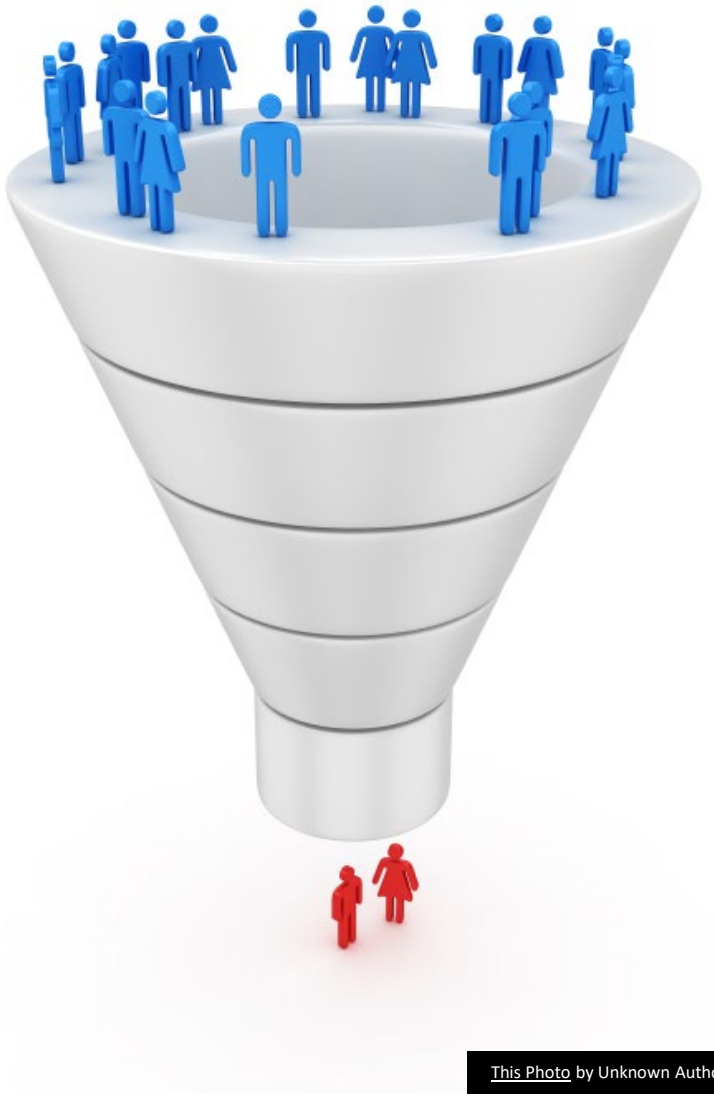
Explore data

- ggplot2 is a R package dedicated to data visualization. It can greatly improve the quality and aesthetics of your graphics, and will make you much more efficient in creating them.
- [Data visualization with R and ggplot2 | the R Graph Gallery (r-graph-gallery.com)]



Assumptions

- All statistical tests have underlying assumptions
- Analysis must meet the assumptions
- Statistics, like all mathematical disciplines, does not infer valid conclusions from nothing.
- Inferring interesting conclusions about real statistical populations almost always requires some background assumptions.
- Statistical assumption – Wikipedia



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Assumptions

- Population -> Sample
  - Sample representative of the population
- Variables
  - Exposure or independent
  - Confounders
  - Mediators, moderators variables
- Outcome variable
- Assumed relationship between outcome and independent variables

Analyze data

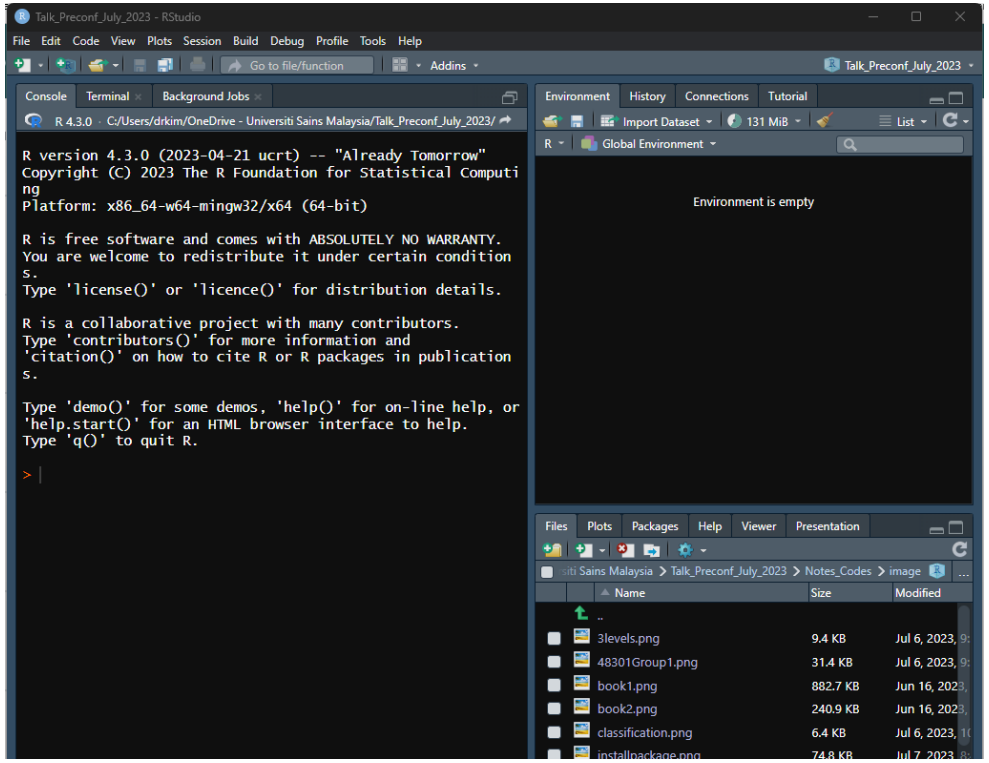


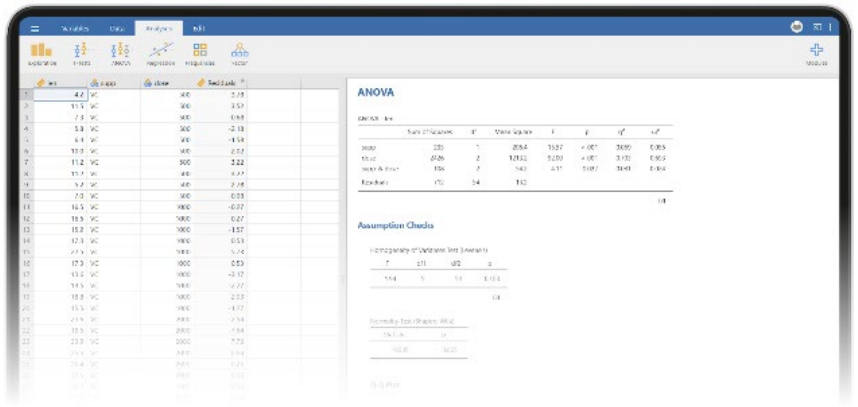
- I use R 95% of the time
- R is a system for statistical computation and graphics.
- R consists of a language plus a run-time environment with graphics, a debugger, access to certain system functions, and the ability to run programs stored in script files.
- R FAQ (r-project.org)
- R IDE
  - https://posit.cloud/content
  - RStudio Desktop – Posit
  - jamovi - open statistical software for the desktop and cloud

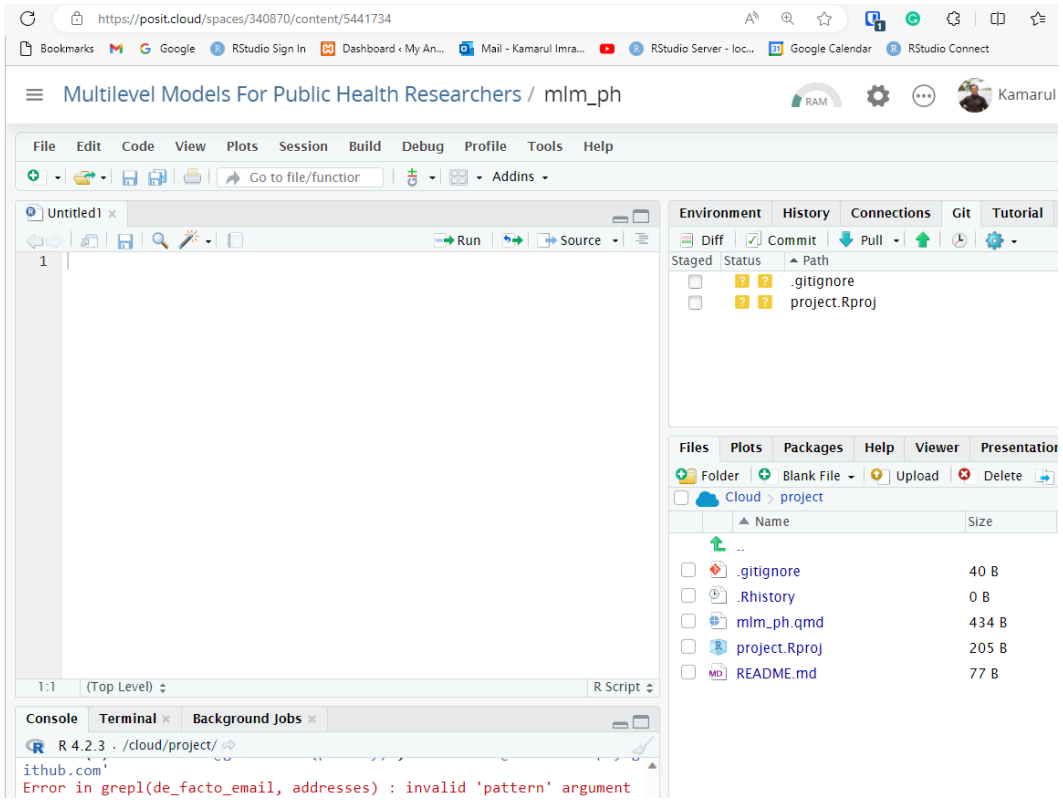




Analyze data : R IDE







Analysis by a team

- Using GIT
- Codes, Results, Reports

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back_up_latex/latex	edited	4 months ago
back_up_references	edited	4 months ago
back_up_yaml	edited	4 months ago
backup_chapter	edited	4 months ago
css	upload	2 years ago
data	remove unused data	10 months ago
figures_for_book	edited	4 months ago
latex	Hanis updated acknowledgement after the t...	9 months ago
plots	edited	4 months ago

Contributors 3

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tengku-hanis Tengku Hanis

wnarifin Wan Nor Arifin

Environments 1

github-pages Active



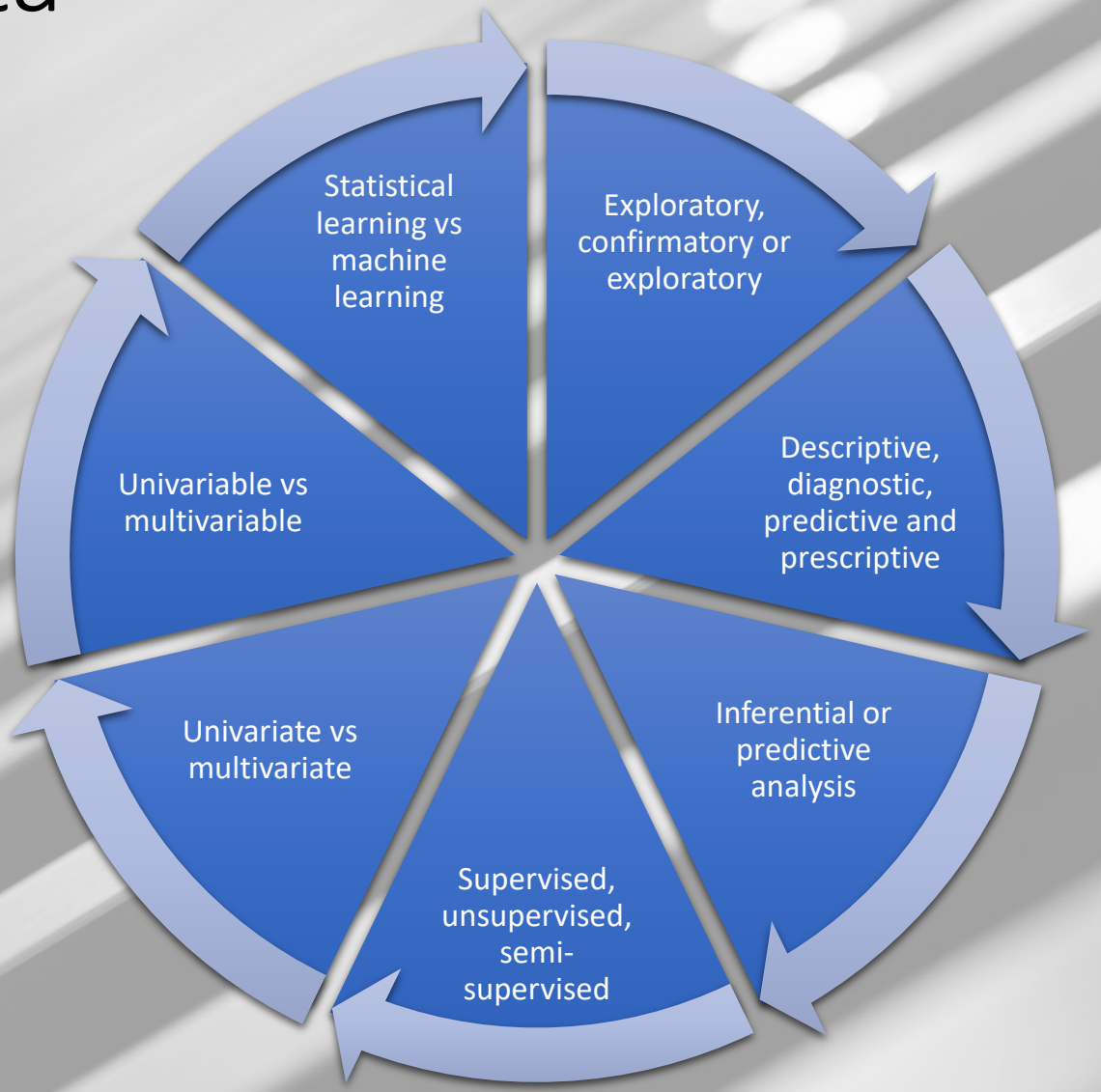
Google Drive

File storage service

OneDrive

File hosting service

Analyze data



Analyze data: Statistical analysis



Estimation



Hypothesis testing



Inference



Prediction

Analyze data: Machine learning (Supervised)



Classification: In classification tasks, the machine learning program must conclude observed values and determine to what category new observations belong.



Regression: In regression tasks, the machine learning program must estimate – and understand – the relationships among variables. Regression analysis focuses on one dependent variable (numerical) and a series of other changing variables – making it particularly useful for prediction and forecasting.



Forecasting: Forecasting is the process of making predictions based on past and present data and is commonly used to analyse trends.

Assess

- Assess assumptions based on model analyzed
  - Make senses?
- Outliers and influentials
- Assess performance
  - R-squared
  - Accuracy
  - Sensitivity, Specificity
- Assess validity
  - Internal validity
  - External validity

Present

- Guidelines
- Basic Statistical Reporting for Articles Published in Biomedical Journals:The “Statistical Analyses and Methods in the Published Literature” or The SAMPL Guidelines”
- SAMPL-Guidelines-6-27-13.pdf (equator-network.org)
- https://www.ahajournals.org/statistical-recommendations

Basic Statistical Reporting for Articles Published in Biomedical Journals: The “Statistical Analyses and Methods in the Published Literature” or The SAMPL Guidelines”

Thomas A. Lang[a] and **Douglas G. Altman**[b]
[a] Principal, Tom Lang Communications and Training International
[b] Director, Centre for Statistics in Medicine, Oxford University



Present

- [American Statistical Association Releases Statement on Statistical Significance and P-Values (amstat.org)]
- “Good statistical practice, as an essential component of good scientific practice, emphasizes principles of good study design and conduct, …
- … a variety of numerical and graphical summaries of data, understanding of the phenomenon under study, interpretation of results in context, complete reporting and proper logical and quantitative understanding of what data summaries mean….

THE AMERICAN STATISTICIAN
2016, VOL. 70, NO. 2, 129–133
http://dx.doi.org/10.1080/00031305.2016.1154108

EDITO

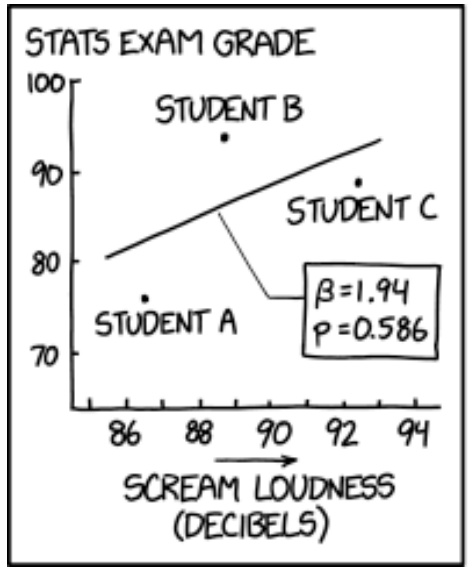
The ASA’s Statement on *p*-Values: Context, P

In February 2014, George Cobb, Professor Emeritus of Mathematics and Statistics at Mount Holyoke College, posed these questions to an ASA discussion forum:

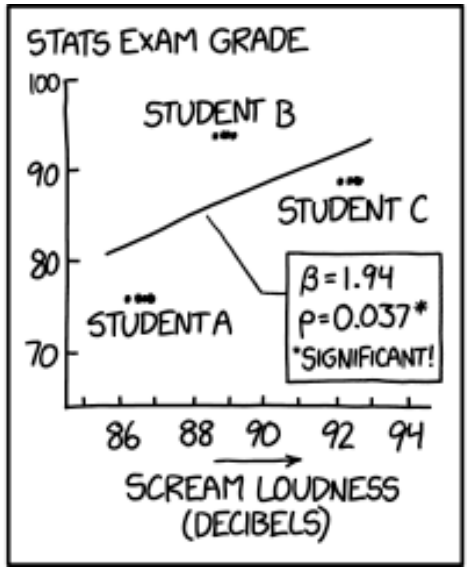
Q: Why do so many colleges and grad schools teach $p = 0.05$?
A: Because that’s still what the scientific community and journal editors use.
Q: Why do so many people still use $p = 0.05$?
A: Because that’s what they were taught in college or grad school.

Cobb’s concern was a long-worrisome circularity in the sociology of science based on the use of bright lines such as $p < 0.05$: “We teach it because it’s what we do; we do it because it’s what we teach.” This concern was brought to the attention of the ASA Board.

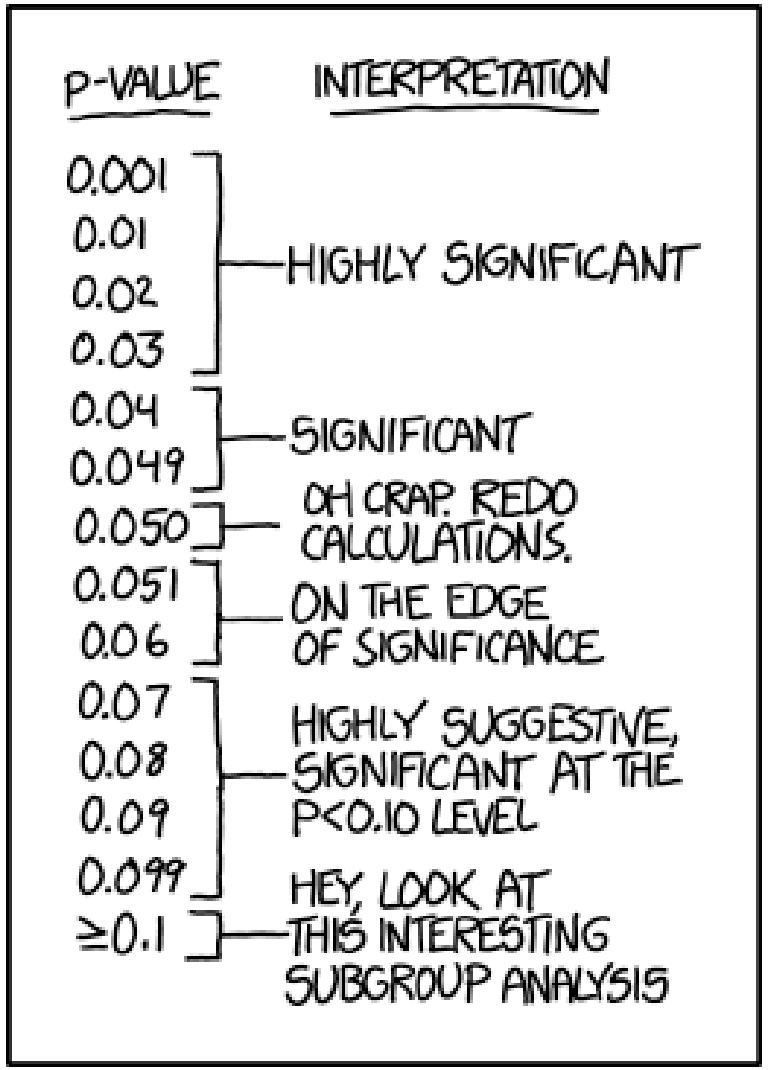
- … No single index should substitute for scientific reasoning.”





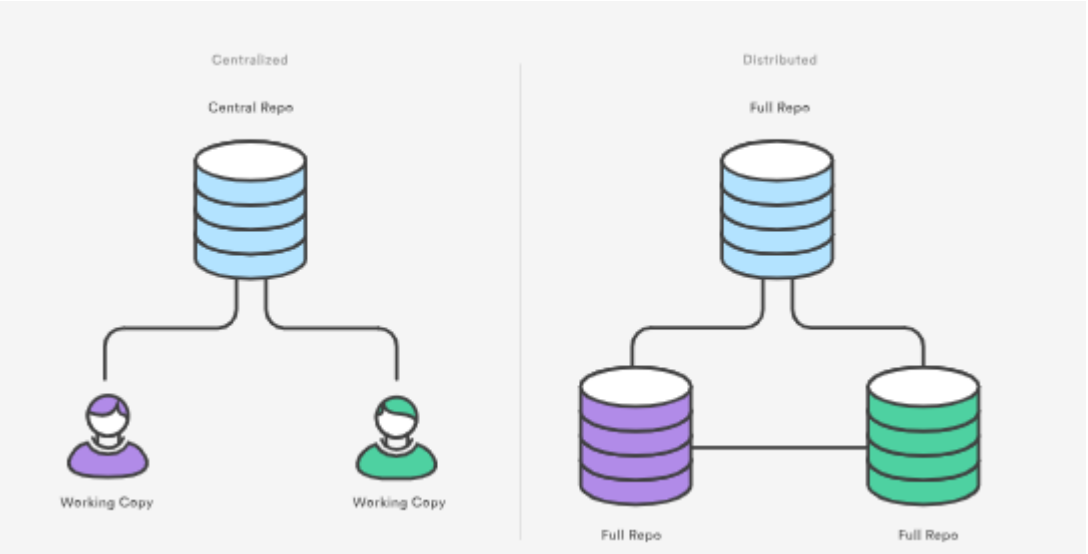


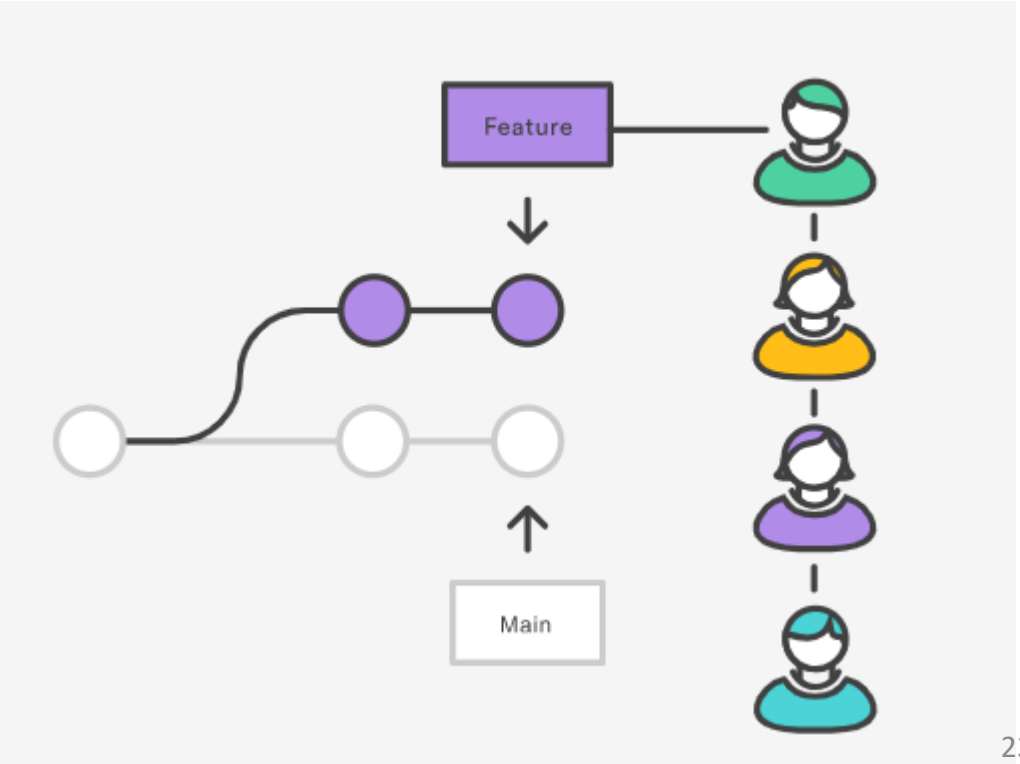




Publish

- Codes
- Store in version control in the cloud
- GitHub, Inc. is a platform and cloud-based service for software development and version control using Git, allowing developers to store and manage their code.
- It provides the distributed version control of Git plus access control, bug tracking, software feature requests, task management, continuous integration, and wikis for every project.
- [GitHub]

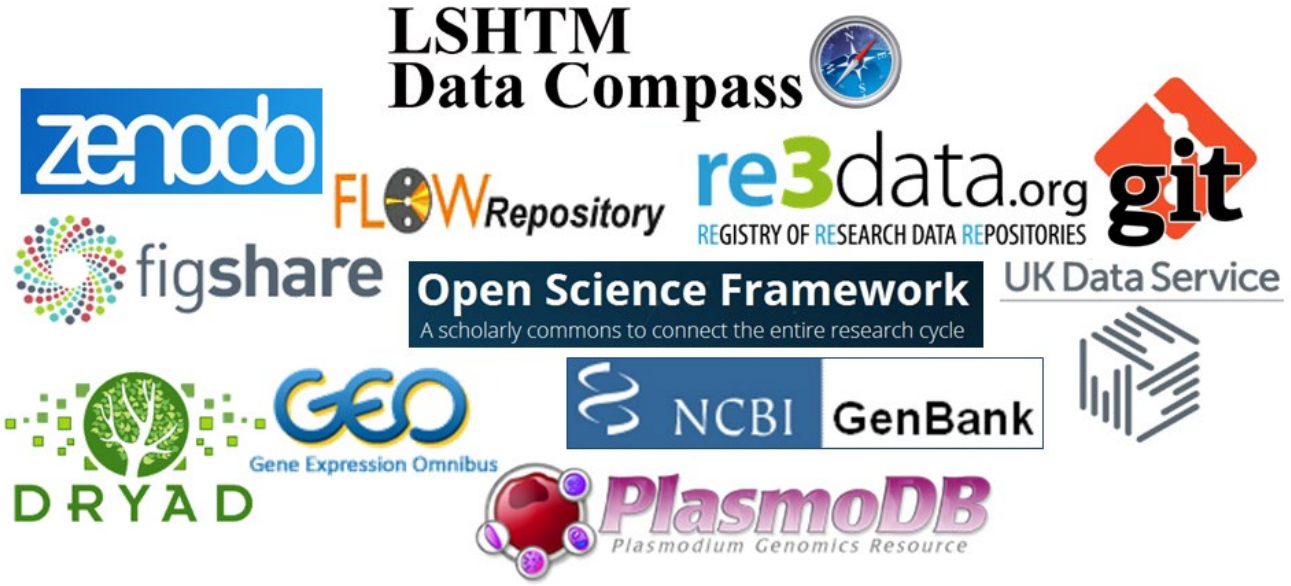




Publish: Data

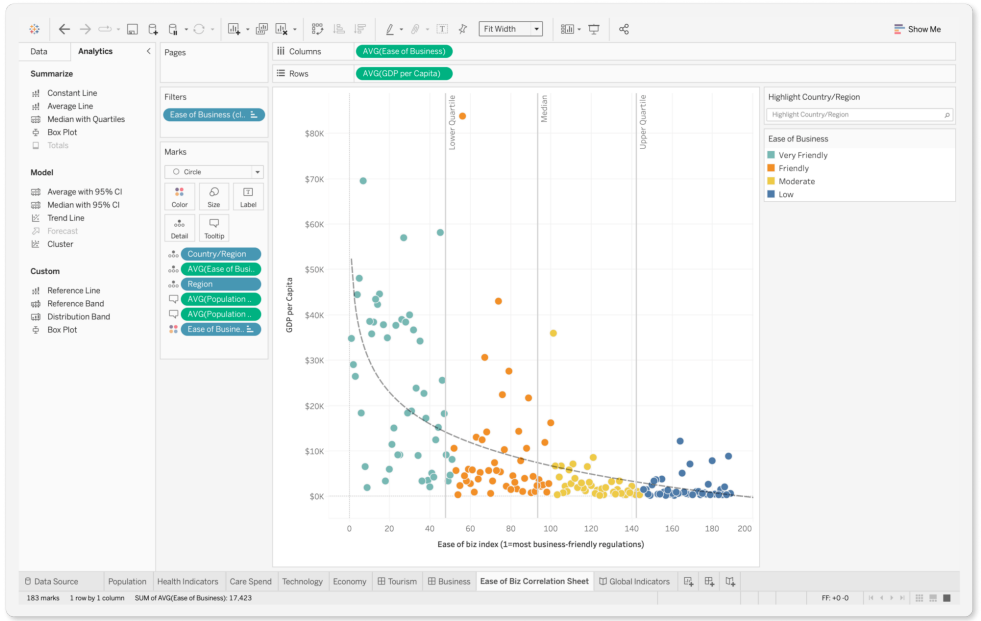
- Credit
- Recognition
- Discoverability
- Reusability
- Awareness
- Accessibility
- Peer-review
- Expert Service
- Technology

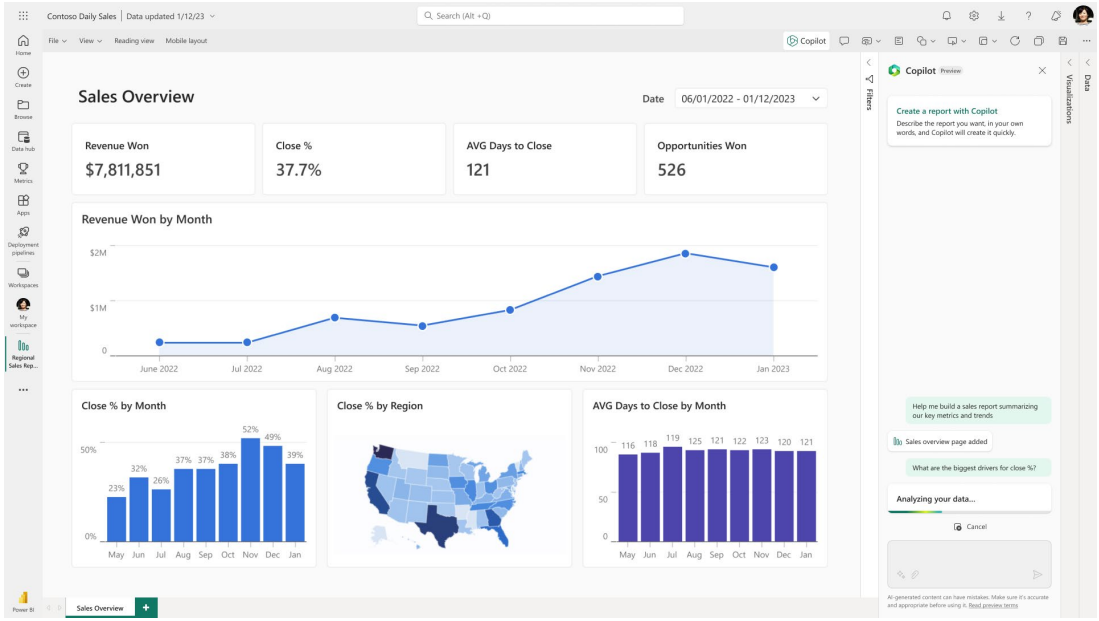




Publish

- Dashboard
  - Shiny
  - Power-BI
  - Tableau





- Shiny
  - Shiny is an R package that makes it easy to build interactive web applications (apps) straight from R. This lesson will get you started building Shiny apps right away.
  - Shiny (posit.co)



- Shiny for R Gallery (posit.co)
- Globorisk for Malaysia (shinyapps.io)



Order Article Reprints

Open Access Article

The Ten-Year Risk Prediction for Cardiovascular Disease for Malaysian Adults Using the Laboratory-Based and Office-Based (Globorisk) Prediction Model

by Che Muhammad Nur Hidayat Che Nawi [1], Mohd Azahadi Omar [2], Thomas Keegan [3], Yong-Poh Yu [4] and Kamarul Imran Musa [1,*]

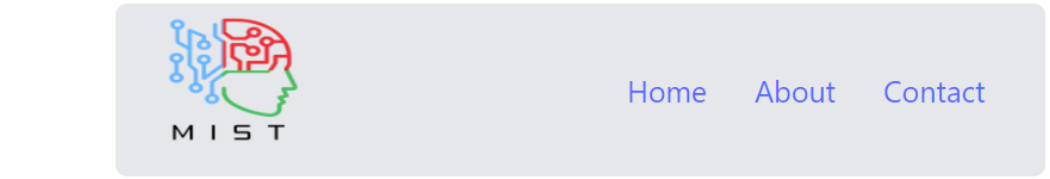
[1] Department of Community Medicine, School of Medical Sciences, Universiti Sains Malaysia, Kubang Kerian 16150, Kelantan, Malaysia

[2] Sector for Biostatistics & Data Repository, National Institutes of Health, Ministry of Health, Shah Alam 40170, Selangor, Malaysia

[3] Faculty of Health and Medicine, Lancaster Medical School, Lancaster University, Lancaster LA1 4YW, UK

[4] Redbeat Academy, AirAsia Berhad, Kuala Lumpur 50470, Selangor, Malaysia

[*] Author to whom correspondence should be addressed.

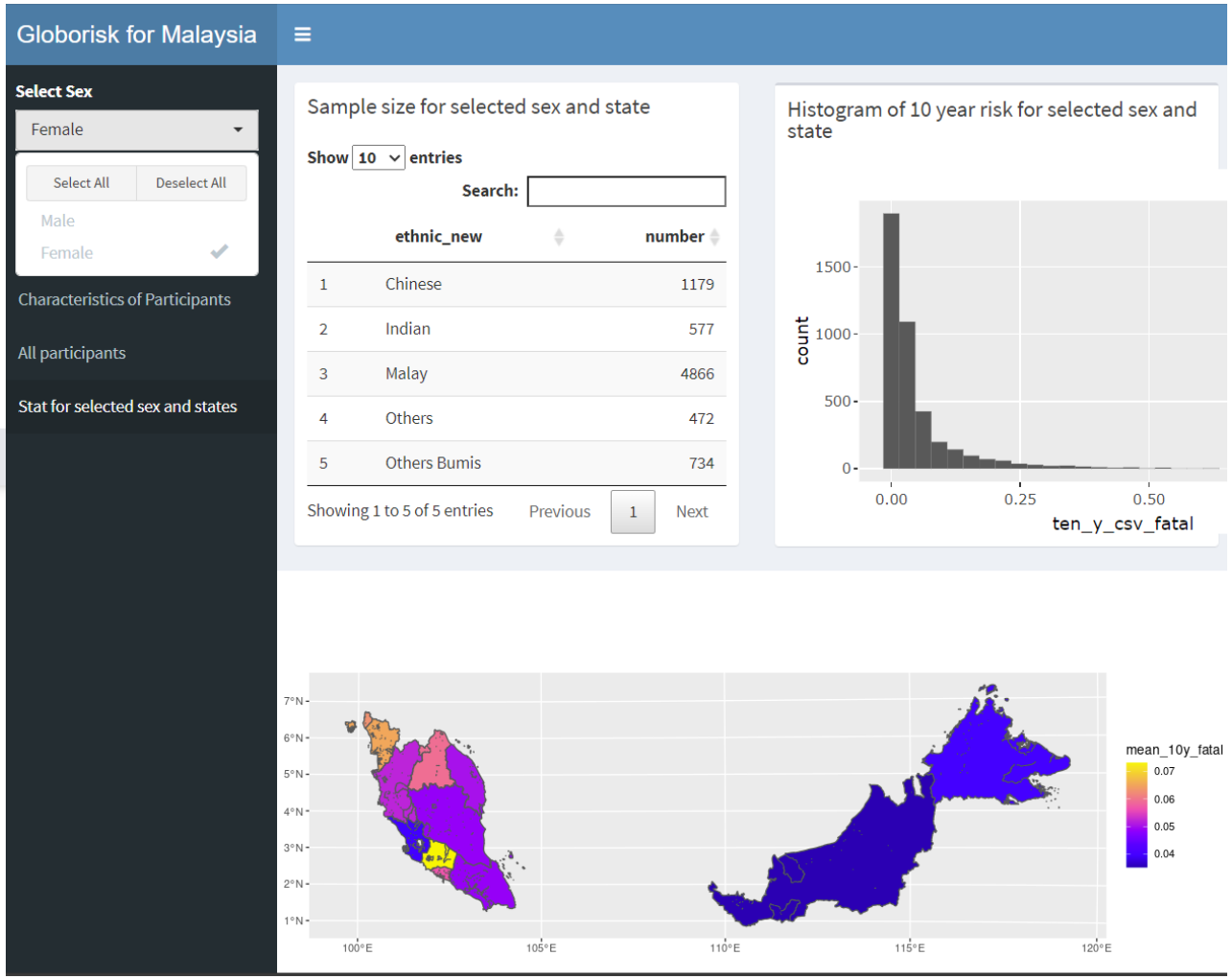


Malaysian Ischemic Stroke Mortality Prediction Tool (MIST)

When to Use | How to Use | Result Interpretation | Disclaimer

Patient Data Form

Age in years:



- mistpredict.com

kim-usm.shinyapps.io/Globorisk4Malaysia/

Books

- [Exploring Data Using R (usm.my)]

Exploring Data Using R







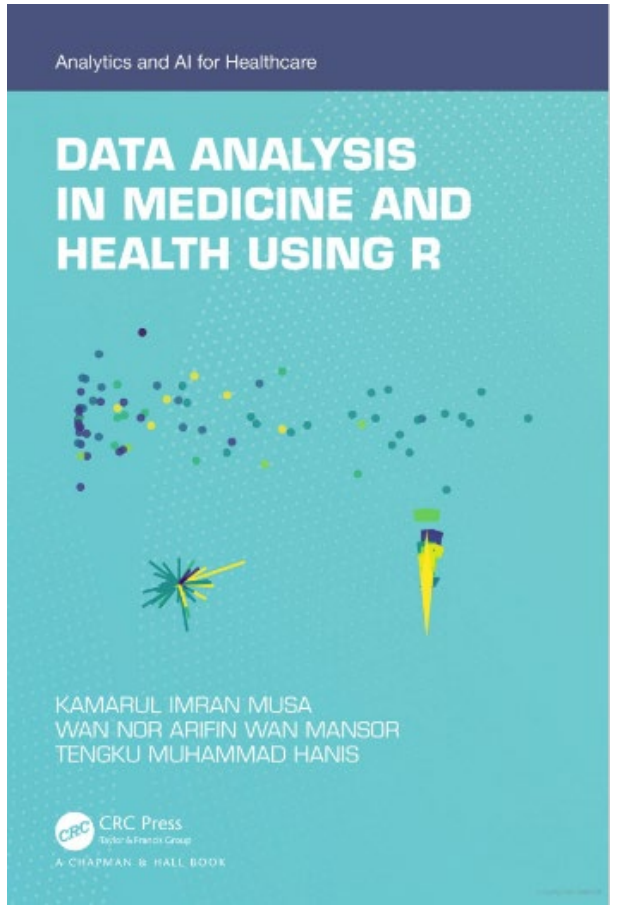






Title: Exploring Data Using R
Authors: Kamarul Imran Musa & Wan Nor Arifin Wan Mansor
Price: RM30.00 **ISBN:** 978-967-461-532-1

- [Data Analysis in Medicine and Health using R - 1st Edition - Kamarul I (routledge.com)]



Summary

- Data are in various formats
- Analysis get more complicated
- Use the right tools to be effective
- Rate your skills in using the tools
- If you want to improve, where do you need to get help?
- You have to work in a team. Leverage others' expertise and 'talent'
- You have to let go off the old tools



Student Guides



Watch Reproducibility and Some Tips for Using Quarto (youtube.com)



Watch Aliran Kerja Analisa Data (Data Analysis Workflow) (youtube.com)



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